Digital Fundamental : DPP of Lec 13

Computer Science

Puneet Ma'am

Q1 Consider the following statements regarding combinational and sequential circuits.
(A) Output of combinational circuits depends on the only current input.
(B) Output of combinational circuit depends on the both current input and previous output.
(C) Output of sequential circuit depends on the current input.
(D) Output of sequential circuit depends on both current input and previous output.
Choose the correct answer from the options given below :
(A) (A) and (C) Only
(B) (A) and (D) Only
(C) (B) and (D) Only
(D) (B) and (C) Only

Q2 Which of the following statements about the types of edge triggering is true?
I. Positive edge triggering occurs on the falling edge of the clock signal, while negative edge triggering occurs on the rising edge.
II. Positive edge triggering occurs on the rising edge of the clock signal, while negative edge triggering occurs on the falling edge.
(A) I is True but II is False
(B) I is False but II is True
(C) Both I and II are True
(D) Neither I nor II is True

Q3 What is one disadvantage of synchronous sequential circuits compared to asynchronous circuits?
(A) They are more complex to design.
(B) They are slower because they depend on clock signals.
(C) They have no feedback paths.
(D) None of the above

Q4 In a synchronous sequential circuit, how is the output pulse related to the clock pulse?
(A) The output pulse is always shorter than the clock pulse.
(B) The output pulse duration is independent of the clock pulse duration.
(C) The output pulse duration is the same as the clock pulse duration.
(D) The output pulse can occur at any time, regardless of the clock.

Q5 A ripple counter is a (n) :
(A) Synchronous Counter
(B) Asynchronous counter
(C) Parallel counter
(D) None of the above

Q6 Which of the following statements is /are true for Sequential circuits?

I: The output depends on both current inputs and previous states.

II: Includes memory elements like flip-flops to store past inputs.

III: These are the combination of logic gates and memory elements.

IV: Adders, subtractors, multiplexers, and demultiplexers are examples of Sequential circuits.

(A) Only I

(B) Only II

(C) Only I, II, and III

(D) Only IV

Q7 Which of the following describes level triggering?

(A) The circuit changes state only on the rising edge of the clock signal.

(B) The circuit changes state based on the clock signal's level rather than its edges.

(C) The circuit ignores the clock signal altogether.

(D) None of the above

Q8 Which of the following are sequential circuits?

A. Flip flop

B. Multiplexer

C. Counter

D. Decoder

E. Adder

Choose the correct answer from the options given below :

(A) B, D Only

(B) A, C Only

(C) C, E Only

(D) D, A Only

Q9 In positive level triggering, when does the circuit change state?

(A) When the clock signal transitions from high to low.

(B) When the clock signal is low.

(C) When the clock signal is high.

(D) At any point during the clock cycle.

Q10 Which of the following best describes synchronous sequential circuits?

(A) They operate without a clock signal and change outputs continuously based on inputs.

(B) They use a clock signal to synchronize changes in output based on input levels.

(C) They are always faster than asynchronous circuits due to their continuous operation.

(D) They require no specific timing considerations for inputs or outputs.

Answer Key

Q1	B
Q2	B
Q3	B
Q4	C
Q5	B
Q6	C
Q7	B
Q8	B
Q9	C
Q10	B

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

Explanation:

- **(A) Output of combinational circuits depends on only the current input**: This is true because combinational circuits produce outputs solely based on the current input values without any memory of past inputs.
- **(D) Output of sequential circuits depends on both current input and previous output**: This is true because sequential circuits have memory, and their production depends not only on the current input but also on the previous output (which is stored in memory elements like flip-flops).

The correct answer is:

Option B: (A) and (D) Only

Q2 Text Solution:

A) Incorrect. This statement reverses the definitions of positive and negative edge triggering.

B) Correct. Positive edge triggering occurs when the clock signal transitions from low to high, and negative edge triggering occurs when it transitions from high to low.

Q3 Text Solution:

A) Incorrect. Complexity can vary and is not solely a disadvantage of synchronous circuits.

B) Correct. The reliance on clock signals makes synchronous circuits potentially slower than asynchronous circuits.

C) Incorrect. Synchronous circuits can include feedback paths; this is not a defining characteristic.

Q4 Text Solution:

A) Incorrect. The output pulse duration matches the clock pulse.

B) Incorrect. The output pulse duration is directly tied to the clock pulse.

C) Correct. In clocked synchronous circuits, the output pulse duration equals the clock pulse duration.

D) Incorrect. Output changes are synchronized to the clock edges.

Q5 Text Solution:

Explanation:

A ripple counter is an asynchronous counter because the flip-flops in the counter are not triggered simultaneously; each flip-flop is triggered by the previous one, causing a delay or "ripple" effect.

The Final Answer is: Option B

Q6 Text Solution:

I: Sequential circuits are those that are dependent on clock cycles and depend on present as well as past inputs to generate any output. The output depends on both current inputs and previous states.

II: Sequential circuits Include memory elements like flip-flops to store past inputs. The output depends upon both the present input and the present state (previous output), so a memory element is required to save the feedback state.

III: Sequential circuits are a Combination of logic gates and memory elements. These circuits are a type of binary circuit that uses logic gates and input variables to process data and make decisions based on past states. They contain memory elements like flip-flops or latches to store binary information from previous inputs.

IV: Adders, subtractors, multiplexers, and demultiplexers are examples of Combinational circuits. So this statement is wrong for Sequential circuits.

Q7 Text Solution:

A) Incorrect. This describes positive edge triggering, not level triggering.

B) Correct. Level triggering depends on whether the clock signal is high or low.

C) Incorrect. The circuit still depends on the clock signal; it just uses its level rather than transitions.

Q8 Text Solution:

Sequential circuits are circuits where the output not only depends on the current inputs but also on the past history of inputs. They contain memory elements such as flip-flops to store information and exhibit sequential behavior.

A. Flip flop: Flip-flops are fundamental memory elements in digital circuits used to store binary data. They are essential in sequential circuits for storing state information.

B. Multiplexer: A multiplexer is a combinational circuit that selects one of multiple input signals and forwards it to a single output based on control signals. It does not inherently store any state information and operates purely based on current inputs.

C. Counter: A counter is a sequential circuit that counts in a predefined sequence and maintains a state. It uses flip-flops to store the current count and has sequential behavior.

D. Decoder: A decoder is a combinational circuit that converts a binary code into a set of outputs where only one of the outputs is active at a time based on the input code. It does not have memory elements and operates purely based on current inputs.

E. Adder: An adder is a combinational circuit that performs addition of binary numbers. It does not store any state information and operates purely based on current inputs.

So option B is the correct answer.

Q9 Text Solution:

A) Incorrect. This describes negative edge triggering.

B) Incorrect. The circuit does not change states when the clock is low.

C) Correct. In positive level triggering, the circuit changes state when the clock signal is high.

D) Incorrect. State changes occur specifically when the clock is high, not at any point.

Q10 Text Solution:

A) Incorrect. This describes asynchronous circuits.

B) Correct. Synchronous sequential circuits utilize a clock signal to synchronize output changes.

C) Incorrect. Synchronous circuits can be slower than asynchronous circuits due to their reliance on clock signals.

D) Incorrect. Synchronous circuits must consider timing for inputs and outputs based on the clock.

Android App | iOS App | PW Website